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31 May 1989
SUPERSEDING
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25 February 1976

MILITARY SPECIFICATION

PRINTED WIRING ASSEMBLIES, PRODUCTION OF

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the production of printed wiring assemblies designed in accordance with MIL-STD-275 and consisting of printed wiring boards on which separately manufactured component parts are mounted.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5999

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SPECIFICATIONS

FEDERAL

O-E-760	-	Ethyl Alcohol (Ethanol); Denatured Alcohol; Proprietary Solvents and Special Industrial Solvents
O-T-236	-	Tetrachloroethylene (Perchloroethylene), Technical
O-T-620	-	1, 1, 1 - Trichloroethane, Technical, Inhibited (Methyl Chloroform)
TT-I-735	-	Isopropyl Alcohol
PPP-B-1672	-	Boxes, Shipping, Reusable with Cushioning
PPP-C-795	-	Cushioning Material, Packaging (Flexible, Cellular, Plastic Film) for Packaging Applications
PPP-C-1752	-	Cushioning Material, Packaging (Unicellular Polyethylene Foam, Flexible)
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-C-1842	-	Cushioning Material, Plastic, Open Cell (for Packaging Applications)

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-117	-	Bags, Sleeves and Tubing
MIL-I-22076	-	Insulation Tubing, Electrical, Nonrigid, Vinyl, Very Low Temperature Grade
MIL-M-38510	-	Microcircuits, General Specification for
MIL-S-45743	-	Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment
MIL-I-46058	-	Insulating Compound, Electrical (for Coating Printed Circuit Assemblies)
MIL-S-46844	-	Solder Both Soldering of Printed Wiring Assemblies
MIL-S-46860	-	Soldering of Metallic Ribbon Lead Materials to Solder Coated Terminals, Process for Reflow
MIL-P-55110	-	Printed Wiring Boards, General Specification for
MIL-C-81302	-	Cleaning Compound, Solvent, Trichlorotrifluoroethane
MIL-B-81705	-	Barrier Materials, Flexible, Electrostatic-Free, Heat Sealable

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts

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MIL-STD-275 - Printed Wiring for Electronic Equipment
MIL-STD-2073-1A - DoD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 1005 - Measurement of Dry-Film Thickness of Organic Coatings Using Micrometers, Standard Test Methods for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ELECTRONIC INDUSTRIES ASSOCIATION

EIA/186-9E - Passive Electronic Component Parts, Standard Test Methods for; Method 9: Solderability

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street NW, Washington, DC 20006.)

THE INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

IPC-A-600 - Acceptability of Printed Boards, Guidelines for
IPC-T-50 - Terms and Definitions for Interconnecting and Packaging Electronic Circuits
IPC-CM-770 - Printed Board Component Mounting
IPC-TM-650 - Test Methods Manual

(Application for copies should be addressed to the Institute for Interconnecting and Packaging Electronic Circuits, 7380 N. Lincoln Avenue, Lincolnwood, IL 60046.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes

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precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Printed wiring assemblies furnished under this specification shall be a product which meets or exceeds all the applicable requirements of this specification and the applicable engineering drawing(s).

3.1.1 Conflict. In the event of any conflict between the requirements of this specification and the applicable engineering drawing(s), the provisions of the engineering drawing(s) shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4. First article samples shall be produced using the same materials and processes as used on production units. Alternatives provided in this specification do not constitute authority to produce production units using different materials or processes other than those used on the first article samples.

3.3 Terms and definitions. Terms and definitions shall be in accordance with MIL-STD-109, MIL-M-38510, and IPC-T-50.

3.4 Material. Material used shall be as specified herein and in the applicable engineering drawing(s).

3.4.1 Materials and processes compatibility. The government provides no guarantee that some combinations of materials and processes specified herein will not produce undesirable side effects detrimental to the end product. It shall be the responsibility of the manufacturer to select those processes and materials which are compatible and produce the desired end product.

3.4.1.1 Cleaning solvents. Solvents to be used for the removal of process contaminants and residues from parts and solder joints shall be compatible with the materials in the parts being cleaned (see 6.4).

3.5 Facilities, tools and equipment.

3.5.1 Tool cleanliness. Tools shall be cleaned prior to use to remove dirt, grease, flux, oil and other foreign matter, and shall be cleaned during use as necessary.

3.5.2 Production facility cleanliness. The supplier is responsible for maintaining assembly areas in a clean, orderly condition. All material not related to the production task should be kept away from work surfaces, and the work area should be swept at least once during each production shift.

3.5.3 Environmental conditions.

3.5.3.1 Assembly and soldering. There shall be no fluxes, acids, solvents, and/or other chemicals allowed in any area in which the assembly and soldering of printed circuit boards is accomplished except as specified in the

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soldering specifications referenced herein. In addition, areas for assembly and soldering of printed circuit boards shall be of controlled access to prevent introduction of unauthorized materials (e.g., on the persons of operators and incidental to other manufacturing processes). In the event printed wiring boards are stored prior to their assembly into printed circuit cards, care shall be taken to ensure their protection. Between final cleaning and application of conformal coating, unfinished printed wiring assemblies shall not be handled with bare hands or visibly contaminated tools. Storage and processing shall take place in a controlled environment having a temperature of 24° ± 6° Celsius (C) (75° ± 10° Fahrenheit [F]), and a relative humidity of 65 percent, maximum.

3.5.3.2 Electrostatic and electromagnetic protection. For electronic components susceptible to electrostatic or electromagnetic damage, special precautions shall be taken by grounding, use of barrier material in accordance with MIL-B-81705, or other appropriate means.

3.6 Production.

3.6.1 Dimensions and tolerances. Dimensions and tolerances shall conform to the applicable drawing(s). Unless otherwise specified, all tolerances are considered absolute and shall be reduced by the amount of the measuring error (see 4.5.1.3 and 4.4.5.1).

3.6.2 Printed wiring boards. Printed wiring boards upon which the components are mounted shall comply with MIL-P-55110 unless otherwise specified on the engineering drawing(s) (see 4.5.1.3 and 4.4.5.1).

3.6.3 Plating. Plating of printed wiring boards shall be in accordance with MIL-P-55110, unless otherwise specified on the engineering drawing.

3.6.3.1 Unacceptable tinning processes. Tinning processes such as unfused electroplated tin/lead, electroless tin (alone), electroplated tin, roller coated tin/lead, etc. shall not be used.

3.6.3.2 Tinning plated printed wiring boards. Tinning shall not be used on gold plated surfaces of printed wiring boards. Surfaces which are plated as specified on drawing(s) shall not be pretinned. Tinning on plated contacts shall be prevented by appropriate masking, using materials which can be cleaned off with approved solvents (see 3.6.6). Solder overlap of the plated surfaces shall not exceed .050 inch.

3.6.3.3 Excess solder on tinned blank boards. The solder coat on fused tin/lead plate or solder dip/level processes shall be continuous and shall be 0.0003 inches thick minimum when measured at the center of the cross section of any applicable conductor, or at the crest (the highest solder point). Any excess buildup of solder coating shall be removed.

3.6.4 Component parts. Component parts which are mounted on the board shall be as specified on the drawing and its parts lists and shall meet or exceed any specifications referenced thereon.

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3.6.4.1 Solderability of component leads and wires. Component leads and wires shall be sufficiently solderable to meet the requirements of this specification including sub-tier specifications cited herein. Gold plated conductors to be soldered shall have the gold plating removed by double dipping using materials specified in MIL-S-45743 or other non-mechanical process prior to assembly. Other leads and wires may be tinned prior to soldering. When required by the contract or purchase order, the solderability shall be tested in accordance with EIA/186-9E or MIL-STD-202, Test Method 208 (see 4.5.1.1).

3.6.4.2 Component lead tinning. Tinning of component leads or wires as necessary to meet the soldering requirements of this specification shall be performed by hot dip or hand soldering and shall comply with MIL-S-45743 (see 4.5.1.1).

3.6.4.3 Component part mounting. Mounting of component parts on the printed wiring board shall be in accordance with the applicable drawing(s), this specification and MIL-S-45743. Component part mounting not covered by these documents shall conform to the requirements of IPC-CM-770 (see 4.5.1.1).

3.6.4.3.1 Wire splicing. Wire splicing shall not be allowed (see 3.6.7.1).

3.6.4.3.2 Wire and lead cutting. Shear type cutters or lead saws shall be used for cutting all semiconductor or shock sensitive device leads. The combination of a cutting and swaging tool when swaging is permitted (see 3.6.4.3.9) may be used provided the leads, after cutting, meet all inspection criteria specified herein.

3.6.4.3.3 Damage to insulation. The conductor shall not be scraped, ringed, nicked, cut, scored, or otherwise reduced in size after stripping. The insulation shall not be compressed, pinched or otherwise reduced in thickness by more than 20 percent of the original nominal insulation thickness. The insulation shall not have gouges, ragged edges, or be loose or frayed. Slight discoloration from thermal stripping is acceptable. Charred insulation is not acceptable. After the insulation is removed, the wire ends of stranded wire shall be twisted in the same direction as the original lay.

3.6.4.3.4 Insulation clearance. Insulation clearance shall be as specified in MIL-S-45743.

3.6.4.3.5 Axial lead component lead forming. The forming action shall be accomplished by lead bending tools, bending the lead ends, as illustrated in Figure 1. Lead bending tools shall be of the type especially designed and commonly sold commercially for this purpose. When being formed, welded leads shall be firmly held by a suitable tool on the side of the weld away from the component body. Lead forming shall not damage the lead due to stressing, nicking, or deformation in excess of the requirements of this specification, and no detrimental energy from the bending action shall be transmitted into the component body. Automatic lead forming devices are acceptable. Smooth impression marks, due to bending tool holding forces necessary to accomplish the bend shall not be cause for rejection.

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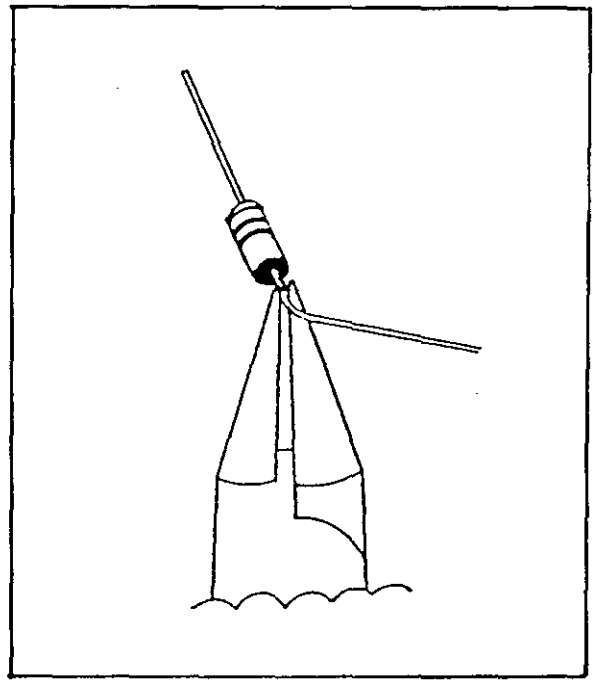


FIGURE 1. Component lead forming.

3.6.4.3.5.1 Stress relief of axial lead component. Leads shall have sufficient slack in the form of a gradual bend, and shall be preformed prior to attachment. When possible, the leads should be formed in the manner shown in Figure 2 or 3, as applicable. Unless otherwise specified on the applicable drawing, the minimum bend radius of the formed lead shall be twice the lead diameter. The minimum distance from the component end seal to the start of the bend shall be .030 inch. For components having a welded lead, such as liquid-electrolyte tantalum capacitors, the start of the bend shall be .030 inch or more from the weld (see Figure 3). When assembling printed wiring assemblies of older design wherein the 0.030 and 2d dimensions will not permit installation of component(s) with vertical leads, the 0.030 dimension may be reduced to not less than 0.015 and the 2d dimension may be reduced to not less than 1d. This reduction is applicable on a component-by-component basis. Where the use of the larger radii are possible, they shall be used even on boards where some leads must be bent to the lesser radii.

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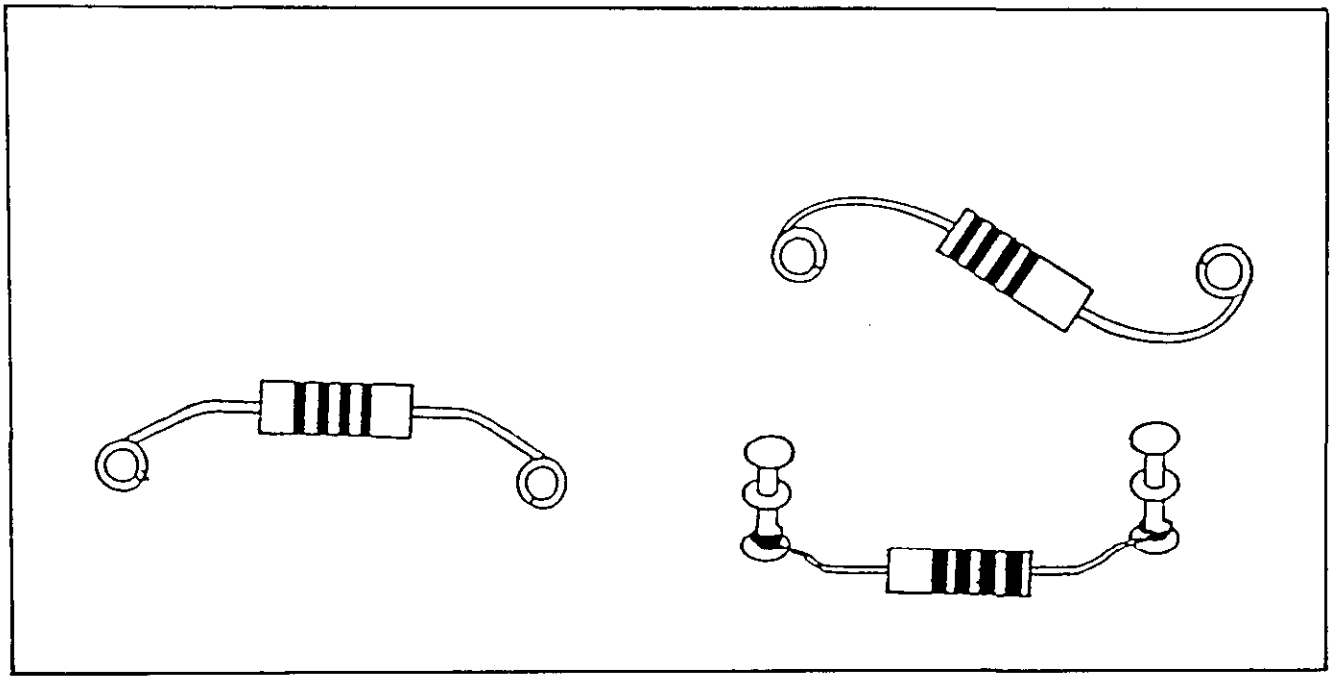


FIGURE 2. Component stress relief.

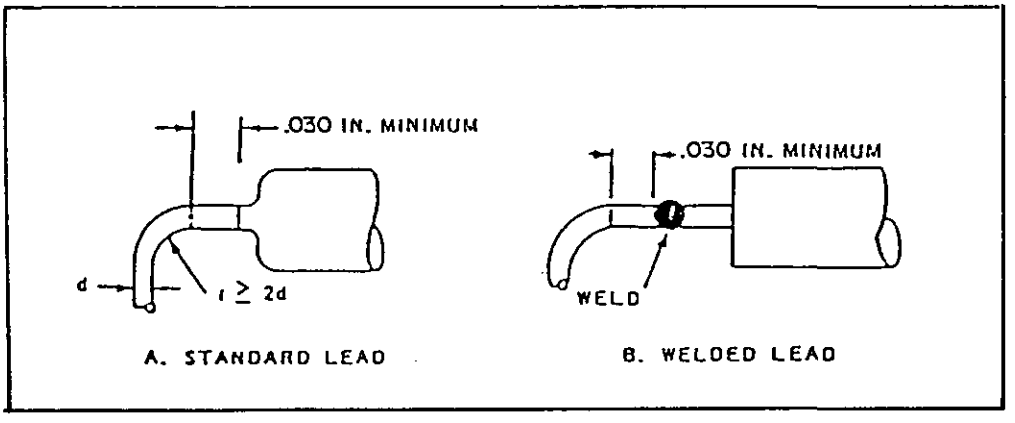


FIGURE 3. Minimum lead bend.

3.6.4.3.5.2 Stress relief of wires to terminals. Wires shall be formed or mechanically connected to the terminal to transmit any stress on the wire to the mechanical connection and not to the solder (see Figure 4). The first wire shall be connected adjacent to the header (nearest to printed wiring board) and each adjacent wire shall be added thereto progressing outward. There shall be maximum practical separation between each adjacent wire on the terminal.

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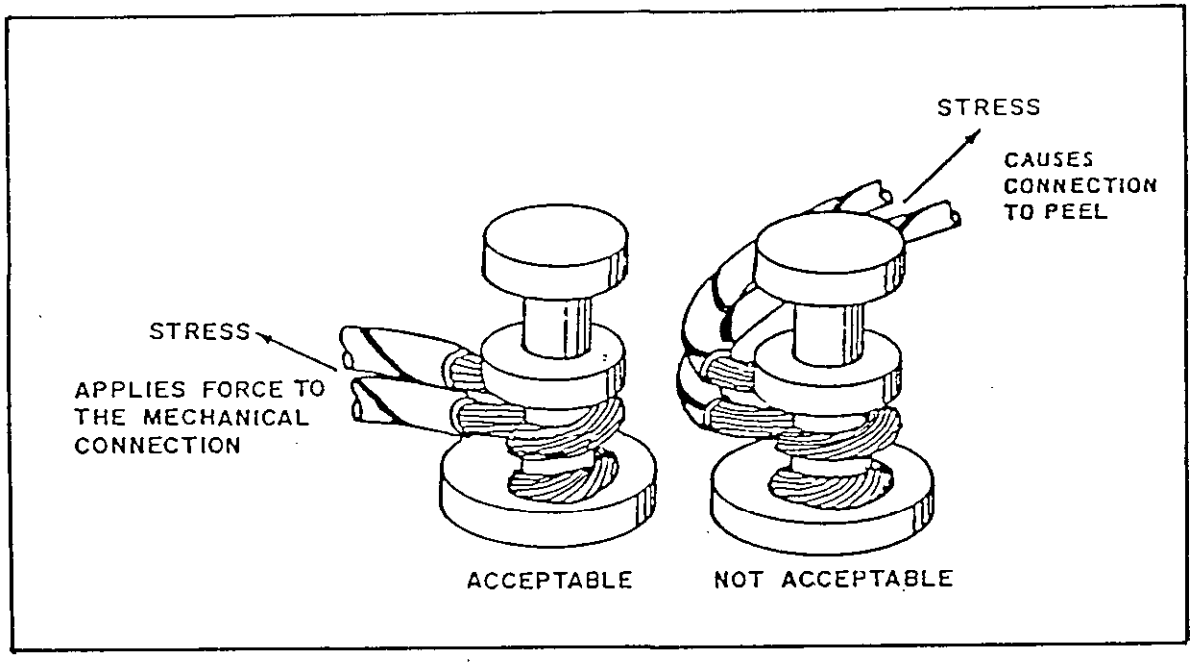


FIGURE 4. Stress transmission to mechanical connection.

3.6.4.3.5.3 Flat pack/ribbon lead components. Part leads shall be formed to a configuration that will prevent detrimental stresses to the solder termination and part seal. The distance from the part lead seal to the first bend shall be 0.020 inch minimum.

3.6.4.3.5.3.1 Flat pack surface mounting. Leads should be uniformly formed. Where the design provides for through-hole mounting holes, surface mounting shall not be used (see 3.6.4.3.5.3.2). When parts are mounted on the same side of the printed circuit board as their solder connection, the angle of the part lead to the pad surface should be 90 degrees (°) but shall not be more than 95° nor less than 45° after soldering. The minimum dimension of the leads after forming and trimming shall be as shown in Figure 5. Provisions to prevent shorting by cutting off unused leads shall be as specified on the drawing. Ribbon leads shall not extend beyond the edge of the terminal pad on either side by more than one third of the lead width nor by more than 0.01 inch on the end, provided spacing to adjacent circuitry is not reduced below the minimum specified on the engineering drawing. When minimums are not specified on the engineering drawing, they may be obtained from the cognizant Government design activity.

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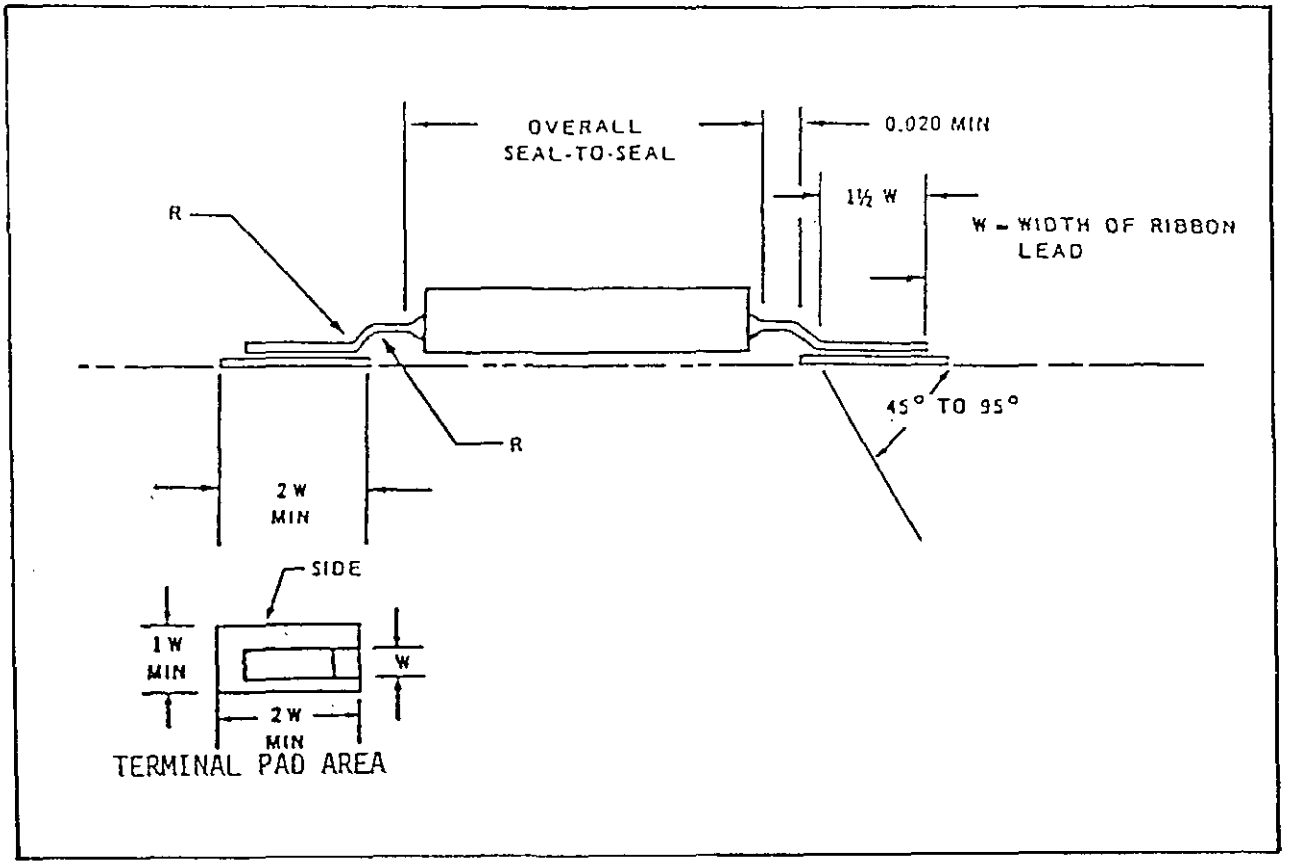


FIGURE 5. Flatpack surface mounting.

3.6.4.3.5.3.2 Flatpack and dual-in-line package (DIP), through-hole mounting. Leads shall be preformed for stress relief. Unless otherwise specified on the drawing, the minimum bend radius of forming shall be twice the thickness of the ribbon lead. The minimum distance from the component body to the start of the bend shall be 0.020 inches minimum (see Figure 6).

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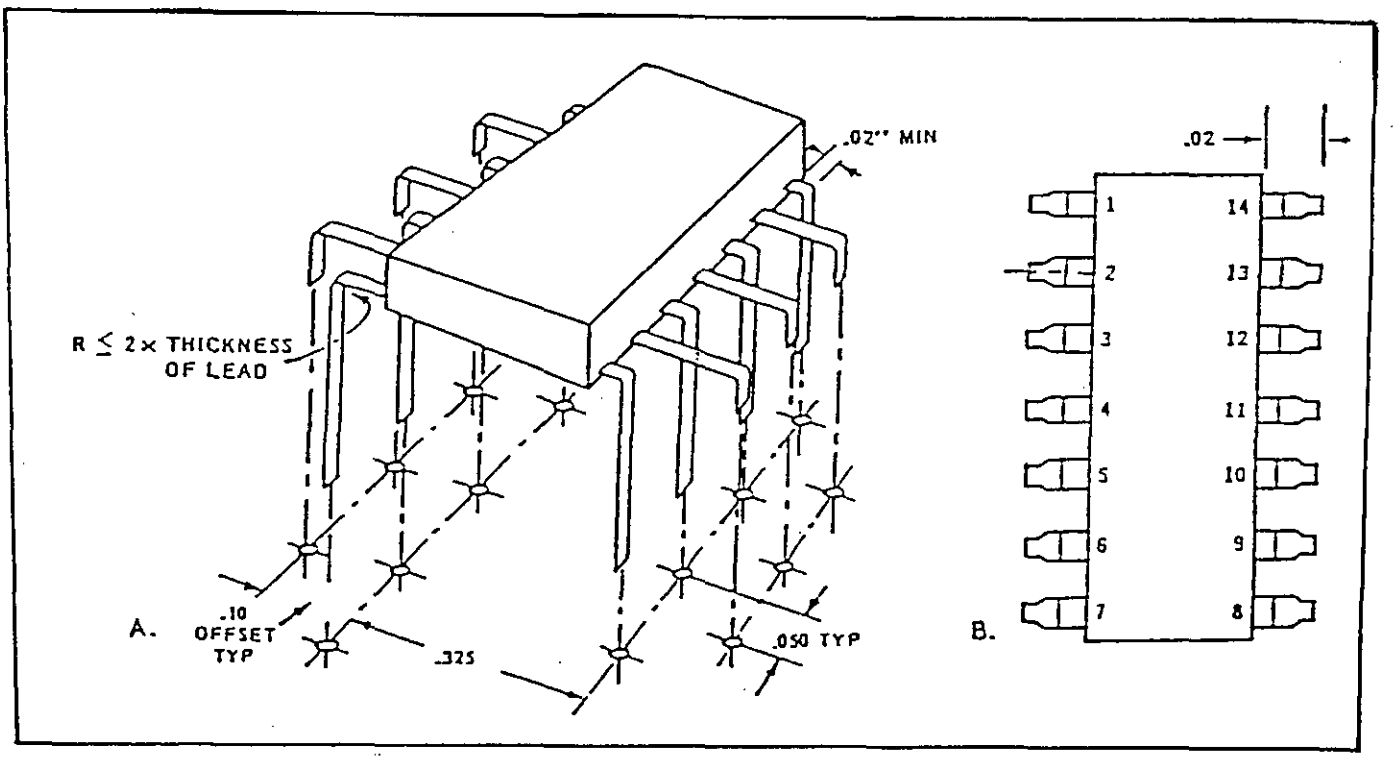


FIGURE 6. Flatpack or DIP through-hole mounting.

3.6.4.3.6 *Component mounting.* *Unless otherwise specified on the* applicable drawings, the components shall be:

a. Centered between termination holes or terminals, as nearly as possible.

b. Axial-lead parts not clamped or otherwise supported shall be mounted parallel to, and in contact with, the printed wiring board. Maximum deviation shall not exceed **0.010** inches above the board at any point. Non-axial and vertically mounted axial lead devices including resistors, capacitors, trimpots, printed wiring board mounted relays, transistor and integrated circuit components with lead patterns matching the printed wiring board patterns and not specifically provided with an additional insulating device for their mounting shall be mounted, where design allows, with their bodies 0.010 to 0.060 inches above the printed wiring board surface.

c. Non-axial lead devices such as transistors and integrated circuit components whose lead patterns do not match the printed wiring board patterns shall have their leads fanned to the printed wiring board pad position. When an additional insulating device which facilitates such fanning is provided for the mounting of the parts, the mount of the combination of parts shall be in accordance with b, above. When such a device is not specified, the bodies of such parts shall be **0.050** to 0.125 inches above the printed wiring board to provide the necessary space for lead fanning.

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d. Preferred positioning is that the part markings shall be visible for inspection. Value markings shall have precedence.

e. When assemblies are to be covered by a rigid epoxy conformal coating, glass encased components (e.g. diodes) shall be protected against thermal expansion by enclosing them in a snug fitting tubing conforming to MIL-I-22076 or as specified on the engineering drawing(s).

f. Metal-cased components that lie over conductors shall be insulated.

g. Lead termination shall be Type I (clinched leads) unless otherwise specified on the drawing. This shall not apply to modules, DIPS or pins. (Pins are considered too large in diameter, or too rigid to be bent.)

h. Components having three or more leads, and polarized components shall be mounted so that their leads terminate in the proper locations according to the engineering drawings.

3.6.4.3.7 Lead termination, Type I (clinched leads) (see Figure 7). Component leads and other conductors terminated directly at the printed wiring board terminal pad shall extend through the board a minimum of one half the pad radius after clinching and a maximum of the largest pad dimension after clinching except as stated below. Unless otherwise specified, the lead bend direction shall be along the conductor pattern, parallel to the board and in contact with the printed wiring. Normal springback after being completely clinched is permissible. Leads shall not extend beyond the terminal pad or printed wiring pattern unless the resultant electrical spacing between adjacent conductors meets the requirements specified in the assembly drawing or the minimum conductor spacing specified in MIL-STD-275. Leads shall point as nearly in the opposite direction as the wiring pattern permits for mechanical retention prior to and during the soldering operation. On circular components with three or more leads, radial orientation of lead pointing shall be used where the wiring pattern permits. When automatic or semi-automatic insertion of components is used with machine clinching of component leads, direction of clinch is optional.

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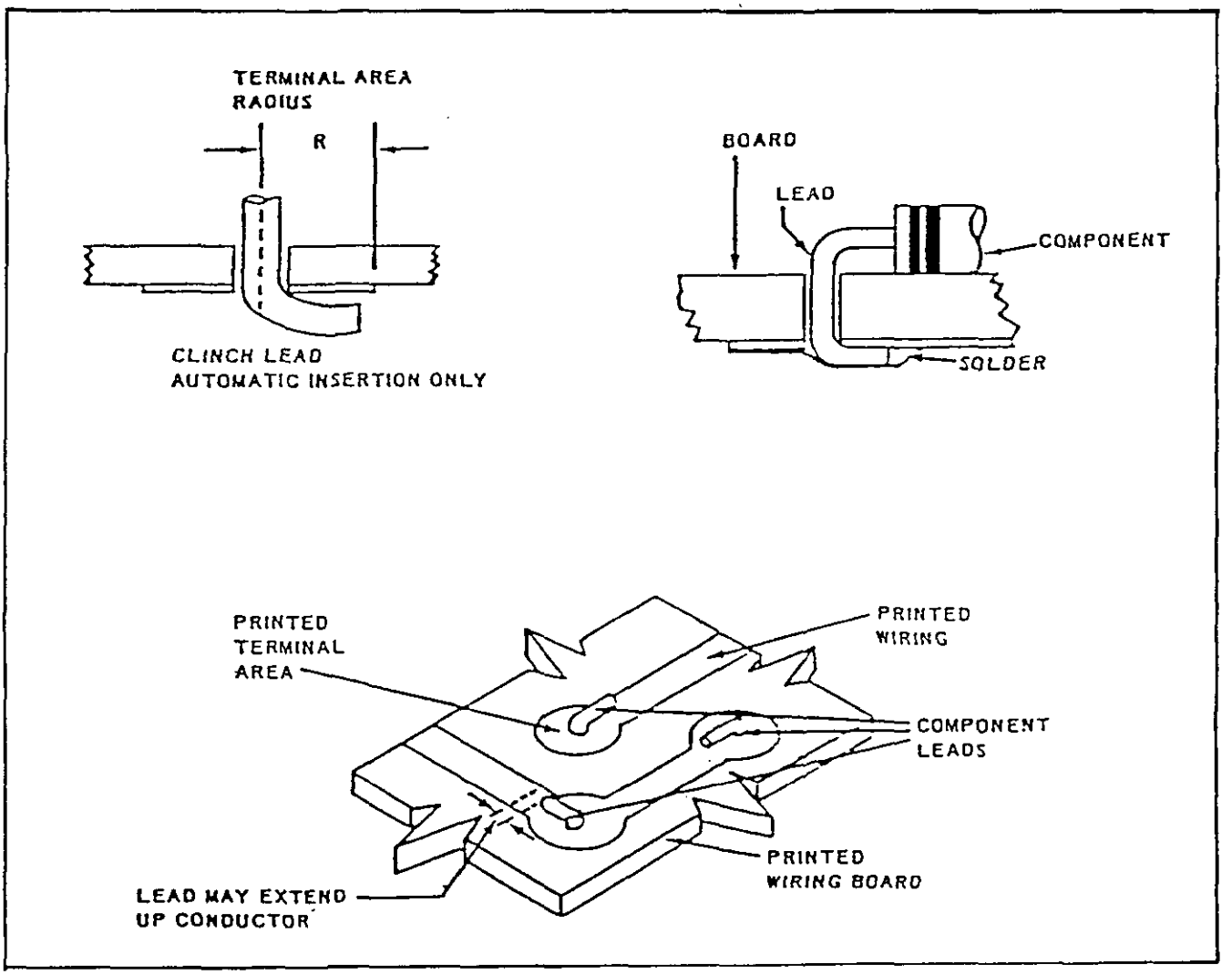


FIGURE 7. Type I lead termination.

3.6.4.3.8 Lead termination, Type II (15° - 45° bend). Component leads shall be bent at an angle to a perpendicular line through the board. Lead length (L) shall be 0.030 (+ 0.030, - 0.00) inch as measured from the conductor surface, the top of an unsupported or plated-through hole, or eyelet (see Figure 8). If annular rings are less than 0.030 inches in width, the lead may extend beyond their outer circumference, provided the resulting electrical spacing between adjacent leads meets the requirements specified in the assembly drawing, or the minimum conductor spacing specified in MIL-STD-275.

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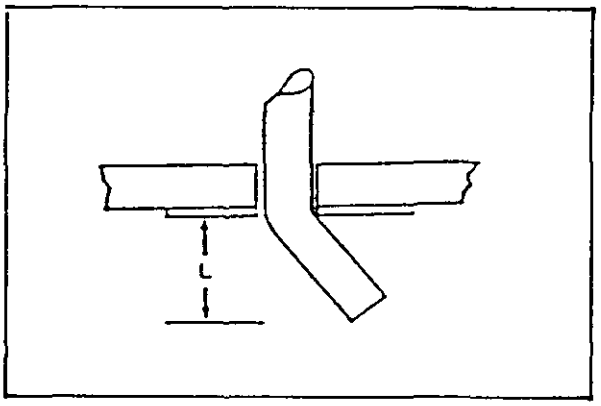


FIGURE 8. Type II lead termination.

3.6.4.3.9 Lead termination, Type III (straight through or swaged). Component leads shall extend through the board and shall be 0.030 (+ 0.030, - 0.00) inch as measured (L) from the conductor surface, the top of an unsupported or plated-through hole, or eyelet (see Figure 9). The lead end may be swaged or otherwise formed for part retention prior to soldering (see Figure 10).

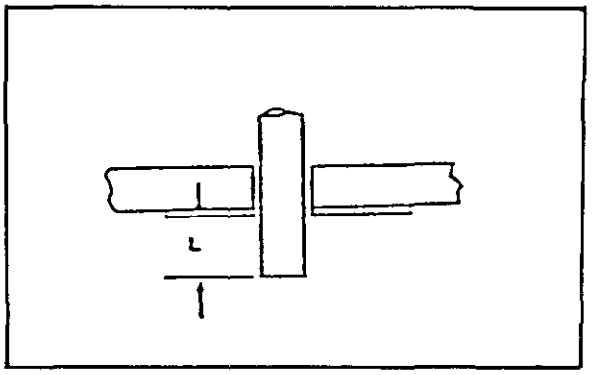


FIGURE 9. Type III lead termination.

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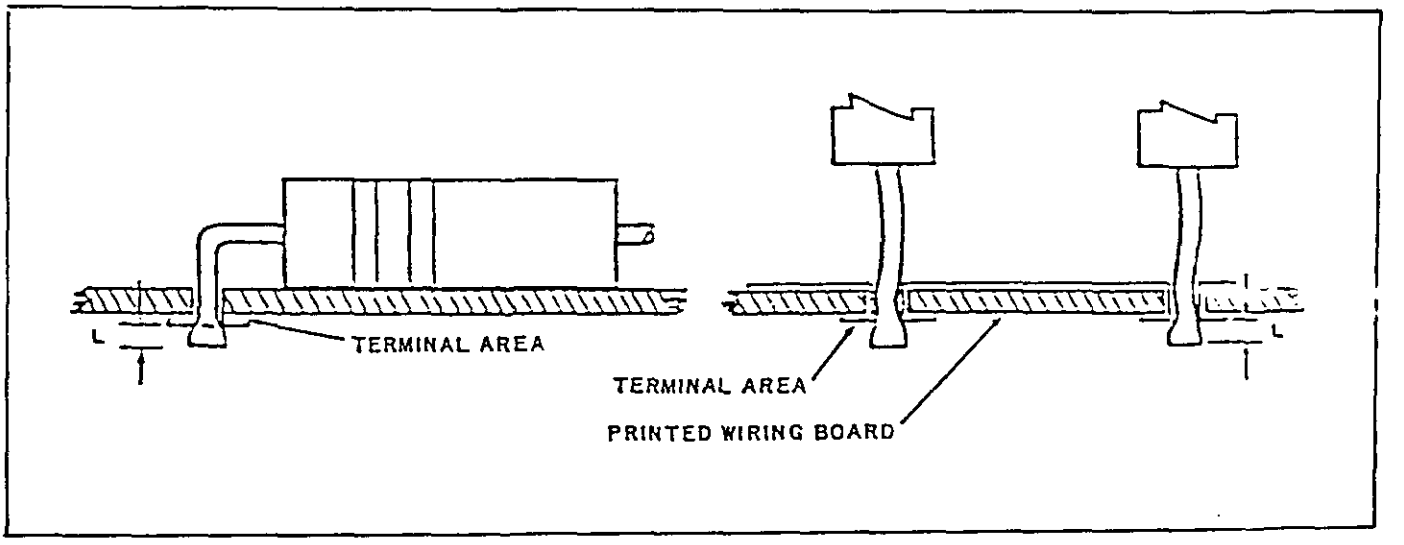


FIGURE 10. Type III swaged leads termination.

3.6.4.3.10 Mounting of terminals on printed wiring boards. Terminals shall be mounted to the printed wiring board as described below (see Figures 11 through 15).

3.6.4.3.10.1 Terminal swaging. Printed terminal areas which will accept terminals shall not be gold plated and shall be tinned. Terminals shall be installed such that the shoulders are in contact with the printed wiring board. A V-type (funnel) swage shall be used on printed wiring boards where the swaged end of the terminal terminates in a terminal area or pattern (see Figure 11). A roll-type swage shall be used only where the swaged end terminates directly to the base material of the printed wiring board (see Figure 14). After swaging, the swaged area shall be free from circumferential splits or cracks, but may have a maximum of three radial splits or cracks, provided the splits or cracks are separated by at least 90 degrees and do not extend beyond the swaged area of the terminal. Flaking or peeling of terminal plating after swaging shall be cause for rejection.

3.6.4.3.10.2 Swaging. When solder rings are specified in the engineering drawing, the ring shall be placed over the terminal shank before swaging (see Figure 12). A V-type (funnel) swage shall be made on the terminal as illustrated in Figure 13. The point of the V-swage shall enter the terminal shank only far enough to produce a finger-tight fit of the terminal.

3.6.4.3.10.3 Terminal soldering. Rolled or flared type terminals shall be soldered to the printed wiring terminal area in addition to the mechanical connection (see Figure 14).

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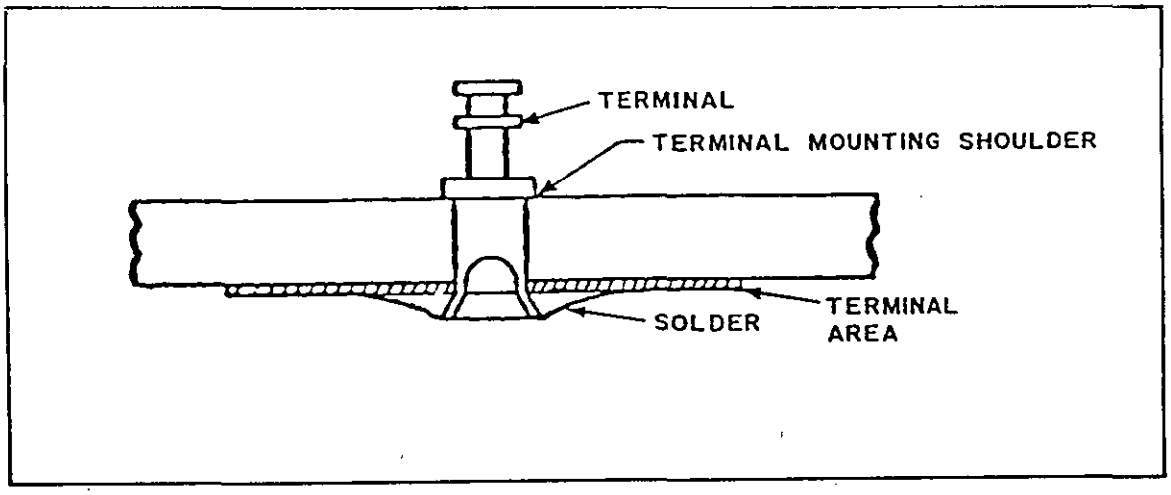


FIGURE 11. Turret terminal installation.

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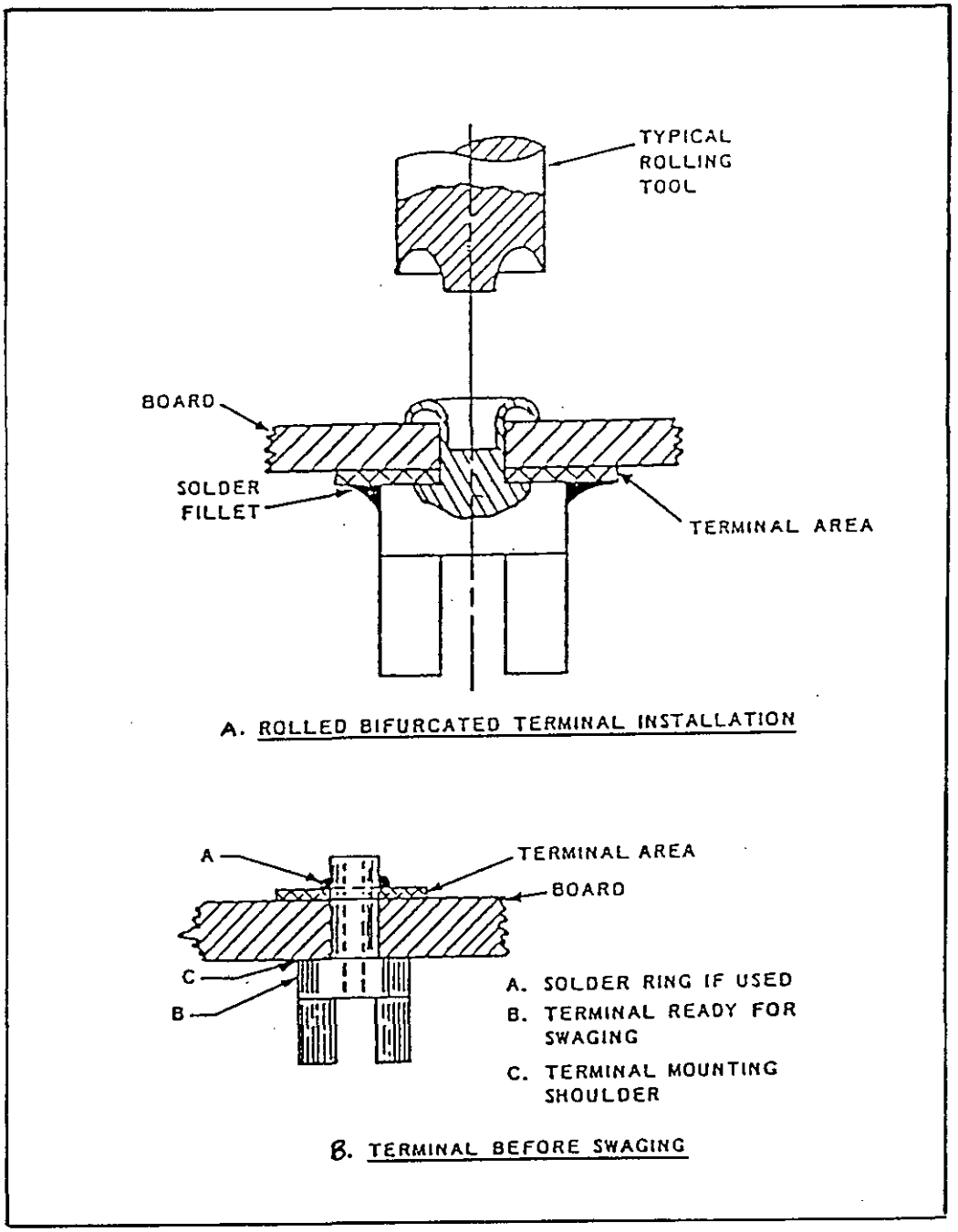


FIGURE 12. Flared terminal installation.

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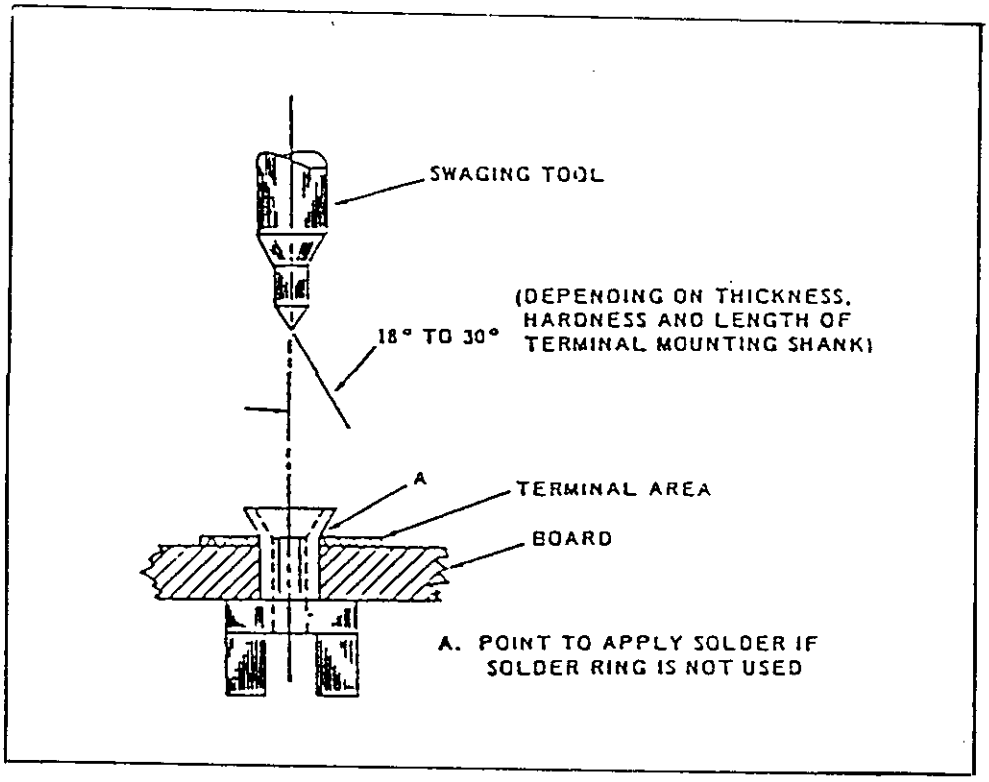


FIGURE 13. Typical "V" type swaging tool and swaged terminal.

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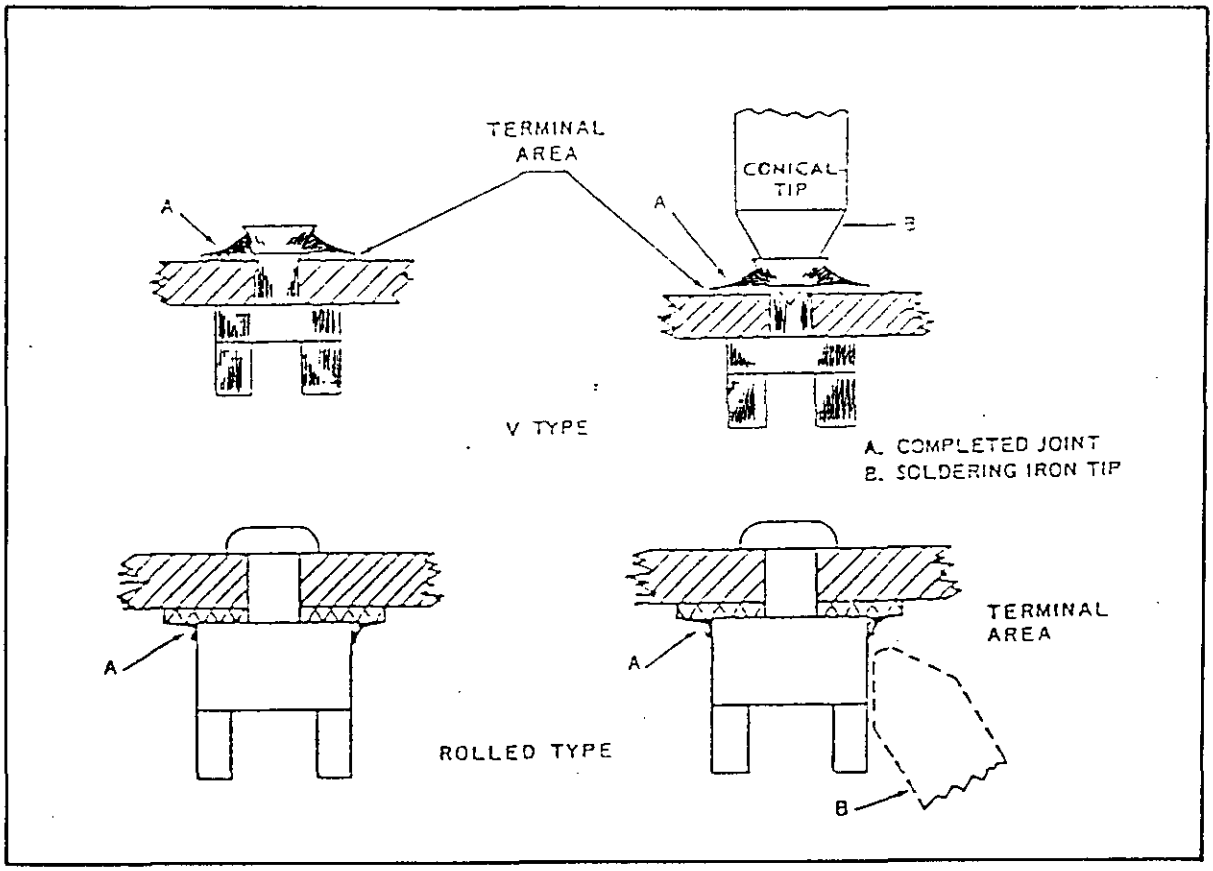


FIGURE 14. Soldering rolled or flared terminals.

3.6.4.3.11 Interfacial connection, printed-wiring boards using plated-through holes. A plated-through hole is not preferred alone as an electrical connection between the printed circuit on each side of the board. The copper lead shall be on both sides of the printed wiring pad (see Figure 15). Solder plugs shall comply with MIL-S-45743 for hand soldering and MIL-S-46844 for machine soldering, except that where components in contact with the printed wiring board preclude their inspection on that side of the assembly, solder plugs shall be considered acceptable if they meet the requirements for the opposite side.

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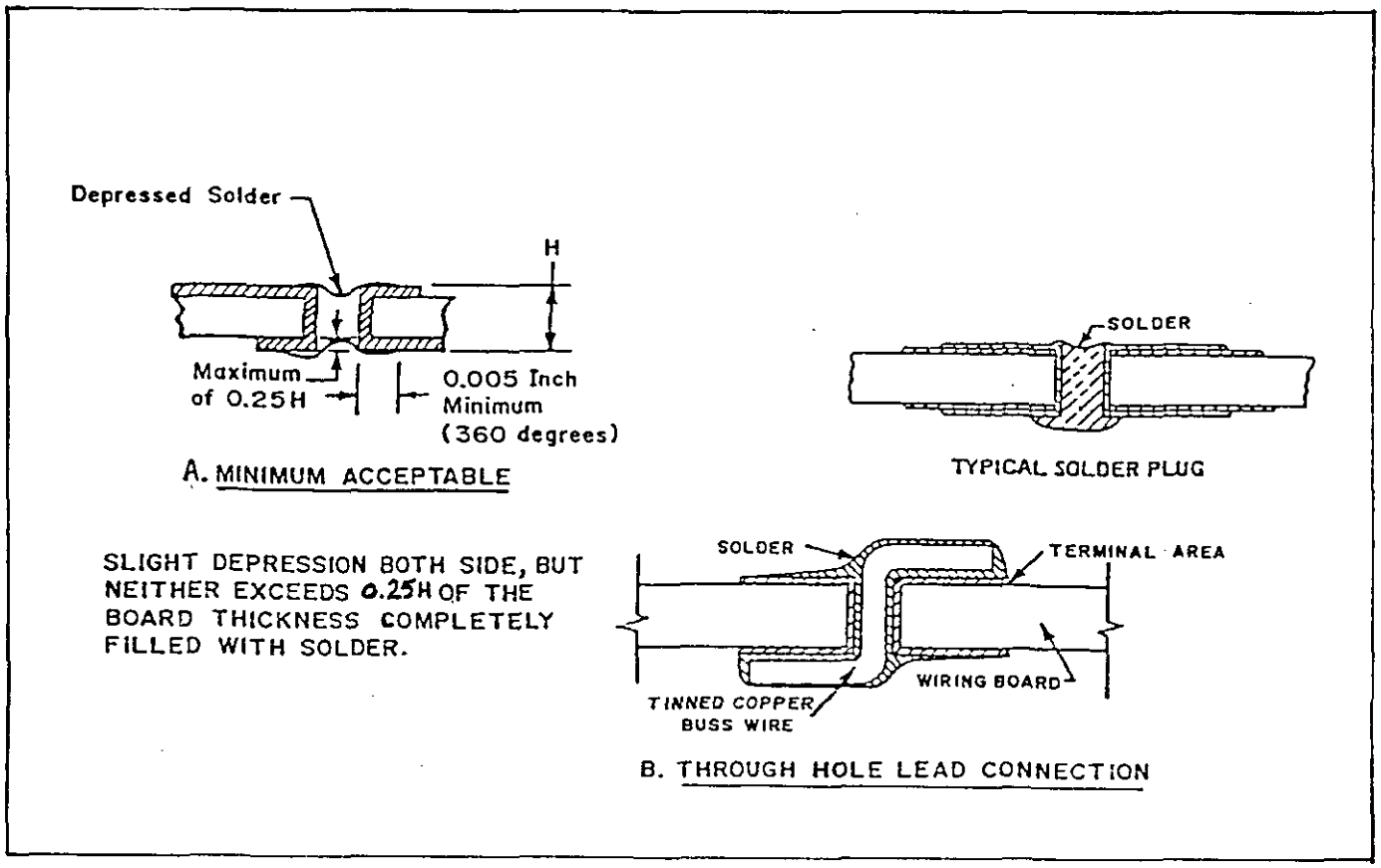


FIGURE 15. Plated through-hole connection.

3.6.5 Soldering. Soldering of components on the printed circuit boards shall comply with the requirements of MIL-S-45743 for hand soldering, MIL-S-46860 for reflow soldering, and MIL-S-46844 for machine soldering, as applicable.

3.6.6 Post-soldering cleaning. Printed wiring assemblies shall be cleaned after soldering using solvents or combinations of solvents which will remove polar and nonpolar contaminants (see 4.5.1.2). Uncoated printed wiring shall be free of flux residues and other contaminants. After cleaning, printed wiring boards shall not be contaminated by handling or environment, prior to conformal coating. When uncoated printed wiring assemblies are tested as specified in 4.5.1.2, the resistivity shall not be less than 2,000,000 ohm-centimeters for each unit of 10 square inches of board surface (counting both sides and the surface areas of the components.)

3.6.6.1 Measling or crazing. Before soldering, measling and crazing shall meet the requirements of IPC-A-600, and shall not exceed 2 percent of

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the printed wiring board area. After soldering and cleaning, measling or crazing shall not bridge between conductors or exceed 3 percent of the printed wiring board surface area. If measling occurs on more than 3 percent of the printed wiring boards, it shall be cause for corrective action. The 3 percent is measured by drawing a line connecting the periphery of all adjacent measling or crazing spots (adjacent is arbitrarily defined as within 0.125 inch) and determining the total area occupied divided by the total area of the printed wiring board. A separate determination is made for each side of the board.

3.6.6.2 Flux residue. After cleaning there shall be no visual evidence of flux residue or other contamination. "Other contamination" includes particles of foreign matter which may result in insulation breakdown or change in electrical characteristics, or degradation of mechanical integrity (i.e., improper bonding of conformal coating).

NOTE: Ultrasonic cleaning may damage certain parts, particularly integrated circuits and semiconductors, and should not be used.

3.6.6.3 Warp or twist. After cleaning (see 3.6.6 and 4.5.1.3), the warp or twist of the printed wiring assembly circuit board shall not exceed 0.010 inches per inch for single and double sided boards and 0.015 inches per inch for multilayer boards unless otherwise specified on the drawing.

3.6.6.4 Final rinse and storage. If deionized water was used as a final rinse after cleaning, and the deionized printed wiring assemblies were not exposed to a controlled time and temperature drying cycle in the cleaning equipment to remove surface moisture, a post bake not exceeding 65.5°C ± 6° (150°F ± 10°) for two hours (± 15 minutes) shall be required to drive off residual moisture before conformal coating. Assemblies shall be properly stored in a dry and dust-free environment between the cleaning and coating processes to prevent moisture contamination.

3.6.7 Repair and rework.

3.6.7.1 Repair of printed wiring conductors and components. Any printed wiring conductor or component which has been broken, cut, cracked or damaged shall not be repaired unless specifically approved by the Government design activity.

3.6.7.2 Rework soldering. Specific areas of soldering found to be unacceptable shall be reworked to conform to the requirements of MIL-S-45743 or MIL-S-46860 as applicable.

3.6.7.3 Defective components. Defective components may be replaced. Soldering removal shall comply with MIL-S-45743. Replacement components and mountings shall meet original specifications.

3.6.8 Conformal coating application. Conformal coating material shall be as specified on the engineering drawing. The material specification and manufacturer's instructions as applicable shall be followed and the material shall be used within the time period specified (both shelf life and pot life).

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When application process information is given in the material specification, it shall take precedence over the manufacturer's instructions and this specification. The coating shall be a minimum thickness (see 3.6.8.5) and fillets shall be kept to a minimum consistent with the size, weight and mounting provisions of the components. A continuous coating shall be applied to both sides of the printed wiring assembly, using a brush, dip, spray or vacuum deposition method, whichever is suitable to the particular assembly and within the following criteria (see paragraph 3.6.8.1 through 3.6.8.8). The coating shall be applied without bubbles, blisters, or breaks which might affect the printed wiring assembly operations, or sealing properties of the conformal coating (see 4.6.4.4.2).

3.6.8.1 Conformal coating fill. After conformal coating, all functional holes shall conform to the dimensional requirements of the applicable engineering drawing(s).

3.6.8.2 Conformal coating on adjustable components. Assemblies having adjustable components shall not have the adjustable portion covered with the coating. Shafts of adjustable components shall be adjusted and sealed in accordance with 3.6.9.2. The junction between adjustable components and the board, or other components of the finished assembly, shall be sealed with the conformal coating material.

3.6.8.3 Conformal coating on connectors. Mating connector surfaces of printed wiring assemblies shall not be coated with the conformal coating. Conformal coating shall, however, provide a seal between the connectors and board and around all mounting devices of the connector.

3.6.8.4 Conformal coating on brackets. Printed wiring assemblies having brackets or other mounting devices shall not have the mating surfaces of said devices coated with conformal coating. The junction between these devices and the board and all attaching hardware shall, however, be coated to provide a seal between the mounting device and the board.

3.6.8.5 Conformal coating thickness. When thickness is not specified on the engineering drawing or material or process specification, the coating thickness for conformal coating (in accordance with MIL-I-46058) for types AR, ER and UR shall be 0.002 ± 0.001 inch; for type SR, the coating thickness shall be 0.005 ± 0.003 inch. For type XY, the coating thickness shall fall within the range of 0.0005 to 0.0020 inch. All thicknesses are measured on a flat, unencumbered surface. For materials other than those specified in MIL-I-46058, the thickness (flat surface) shall be as specified in the material specification or separate process specification when specified on the engineering drawing.

3.6.8.6 Conformal coating on flexible leads. Components which are electrically connected to the printed wiring assembly by flexible leads shall have the junction of these leads with the component and the printed wiring assembly sufficiently coated to provide protection (bend relief).

3.6.8.7 Perimeter coating. The outer perimeter of printed wiring assemblies shall not be increased in total thickness by more than 0.040 inch

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as a result of conformal coating. The "outer perimeter" is defined as the area on each side of the board a distance of not more than 0.25 inch inward from the outer edge.

3.6.8.8 Edge coating. The dimensions of the printed wiring assemblies shall not be increased in length or width by more than 0.030 inch on each edge (total 0.060) by application of conformal coating.

3.6.9 Adjustable components.

3.6.9.1 Adjustment. When required, printed wiring assemblies shall be calibrated or adjusted per drawing requirements after conformal coating.

3.6.9.2 Sealing of adjustable components. After calibration adjustment of components, the movable shafts or other adjustable features shall be spot sealed with a gloss white quick drying cellulose lacquer sealant unless otherwise specified on the drawing. Sealant shall be used unthinned and shall be applied sparingly by brush or applicator to limit penetration to the minimum practical (see 4.4.5.1).

3.6.10 Marking. Marking shall be in accordance with the engineering drawing(s) (see 4.4.5.1).

3.6.11 Functioning of printed wiring assemblies. Upon completion of final assembly the printed wiring assemblies shall function electrically in accordance with the requirements of applicable engineering drawings, test specifications and all other operational requirements (see 4.4.5.2).

3.7 Workmanship. Printed wiring assemblies shall be uniform in quality and appearance. They shall be clean and free of dirt, foreign matter, oil, fingerprints, corrosion, salts, flux residues, and contaminants. All features, conductors, patterns, interfacial connections, jumpers, etc., shall be in accordance with the assembly drawing or printed wiring board master drawing(s) referenced therein (see 4.4.5.1 and 4.6.4.4.2).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the

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specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained.

4.1.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the "General Requirements" of MIL-STD-202.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Component and material inspection (see 4.3)

b. First article inspection (see 4.4 and 6.3)

c. In-process inspection (see 4.5)

d. Quality conformance inspection (see 4.6).

4.3 Component and material inspection. The contractor is responsible for insuring that components and materials used for fabricating printed wiring assembly(s) are in accordance with requirements listed in table I and the referenced documents for providing defect-free components and materials in the printed wiring assembly(s).

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Table I. Component and material inspection.

Component and Materials	Requirement Paragraph	Applicable Specification
Printed Wiring Board	3.4, 3.6.2	Master Drawing, MIL-P-55110
Components	3.4, 3.6.4	As specified on Master Drawings
Plating	3.6.3	Master Drawing, MIL-P-55110
Solder and Soldering Flux	3.6.4.2, 3.6.5	See applicable Soldering Specification(s)
Conformal Coating	3.6.8	Master Drawing, MIL-I-46058
Solvents	3.6.6	O-T-236, O-E-760, O-T-620, TT-I-735, MIL-C-81302
Sealant	3.6.9.2	Master Drawings

4.4 First article. The first article inspection shall consist of the inspections and tests specified in table II in the order shown (see 4.4.2), at a laboratory designated by the Government (see 6.2). Quantity production shall be withheld until the first article sample has been found satisfactory by the procuring activity.

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Table II. First article inspection.

Examination or Test	Requirement Paragraph	Method Paragraph
1. Components and Materials	(See table I)	4.3
2. Visual and Dimensional Examination	Master Drawings and this specification	4.4.5.1
a. Component Mounting	3.6.4.3 thru 3.6.4.3.11, 3.7	4.4.5.1
b. Printed Wiring Board	3.6.1, 3.6.2, 3.6.3 3.6.6, 3.7	4.5.1.3, 4.4.5.1
c. Soldering	3.6.5, 3.7	4.4.5.1
d. Plating	3.6.3, 3.7	4.4.5.1
e. Cleaning	3.6.6 thru 3.6.6.4, 3.7	4.5.1.2, 4.4.5.1
f. Conformal Coating	3.6.8 thru 3.6.8.8, 3.7	4.4.5.1, 4.6.4.4.2
g. Sealing of Adjustable Components	3.6.9.2, 3.7	4.4.5.1
h. Marking	3.6.10, 3.7	4.4.5.1
i. Preparation for delivery	Section 5	4.6.4.4.3
3. Electrical Parameters	3.6.11	4.4.5.2
4. Final Electrical	3.6.11	4.4.5.2

4.4.1 First article sample. The first article sample(s) (see 3.2 and 6.3) shall consist of printed wiring assemblies that are representative of subsequent production. The test specimens shall be fabricated from the same components and materials, and by the same processing procedures (e.g., fluxing, cleaning, soldering) that will be used for production units.

4.4.2 Inspection routine. The samples shall be subjected to the inspections specified in table II, in the order shown. In addition to the

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first article test data, marked-up drawings of the assemblies tested shall be furnished to the Government when a first article sample is required by the Government and indicate the point or location where each type of electrical measurement was made (see 4.4.5.2).

4.4.3 Rejection. One or more failures of any sample may be cause for rejection of the first article sample.

4.4.4 First article approval. Approval of the first article in no way relieves the contractor of responsibility for complying with all requirements of the specifications, referenced drawings and all other terms and conditions of the contract nor shall an approved first article be construed as altering or taking precedence over any of these requirements.

4.4.5 Methods of examination and test.

4.4.5.1 Visual and dimensional examination. Visual and dimensional examination shall be made under 4X to 10X magnification. Higher magnification may be used to inspect questionable defects. Physical dimensions and markings shall be verified in accordance with applicable requirements (see 3.6.1 and 3.6.2).

4.4.5.2 Electrical parameters. The electrical parameters for the printed wiring assembly shall be checked as specified in the engineering drawings.

4.5 In-process inspection. In-process inspection shall consist of the examinations and tests listed in table III.

Table III. In-process inspections.

Sample Plan	Test	Requirement Paragraph	Method Paragraph
4.5.1.1.1	Solderability of Component Leads	3.6.4.1	4.5.1.1
4.5.1.2.1	Printed wiring assembly Cleanliness Test by Resistivity of Solvent Extract	3.6.6	4.5.1.2
4.5.1.3.1	Warp or Twist	3.6.6.3	4.5.1.3

4.5.1 Methods of examination and test.

4.5.1.1 Inspection of components. Prior to release into the production process, if required by contract or purchase order (or if required to meet the solderability requirements of this specification including subtier solder specifications), each lot of components shall be inspected for solderability (see 3.6.4.1) in accordance with EIA/186-9E or MIL-STD-202, Method 208.

4.5.1.1.1 Sampling plan for solderability test. Lot sampling shall be in accordance with MIL-STD-105 special inspection level S-4 with an acceptable quality level (AQL) of 2.5.

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4.5.1.1.2 Inspection lot. An inspection lot shall consist of all components of a part number (1) procured at one time from one source, or (2) that will be issued to production at one time, or (3) in storage in essentially the same condition that will be drawn from for use in production. Sampling may be taken from lots formed in any of the three preceding ways.

4.5.1.1.3 Rejected lots. If an inspection lot is rejected, the supplier shall withdraw the lot and take corrective action by screening out the defective components and replacing or reworking them, as applicable, and resubmitting the lot. Resubmitted lots shall be inspected using tightened inspection in accordance with MIL-STD-105.

4.5.1.2 Ionic or resistivity of solvent extract. Printed wiring assemblies shall be tested in accordance with Method 2.3.26 of IPC-TM-650. Other methods shall have prior approval of the procuring activity.

4.5.1.2.1 Sampling. Immediately after cleaning of printed wiring assemblies, subsequent to soldering of components thereon and any necessary rework, samples shall be selected from each lot and subjected to the tests in table III. Samples shall be selected in accordance with MIL-STD-105, special inspection level S-4 with an AQL of 2.5.

4.5.1.2.2 Rejection. If the sample fails to pass the test specified in 4.5.1.2, the lost shall be rejected.

4.5.1.2.3 Rejected lots. When a lot is rejected as a result of a failure to pass the resistivity of solvent extract test, the supplier shall withdraw the lot, take corrective action with the cleaning materials and procedures, reclean the lot, and resubmit the lot for test in accordance with paragraph 4.5.1.2.1 above. Resubmitted lots shall be inspected using tightened inspection in accordance with MIL-STD-105.

4.5.1.3 Warp or twist (see 3.6.1 and 3.6.6.3). Warp shall be measured as the maximum deviation of the surface of the board from a plane passing through the four surface corners of a printed wiring board.

$$W = \frac{A}{B}$$

A is deviation from plane and B is the length of the board. Twist shall be measured as the maximum deviation of the surface of the board from a plane passing through three surface corners of the printed wiring board.

$$T = \frac{A}{C}$$

A is the maximum deviation of the printed wiring board surface from the plane and C is the diagonal distance from corner to corner of the board.

4.5.1.3.1 Sampling. Immediately after cleaning of printed wiring assembly, subsequent to soldering of components thereon and any necessary rework, but prior to conformal coating, a sample shall be selected in accordance with MIL-STD-105, special inspection level S-4 with an AQL of 2.5.

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4.5.1.3.2 Inspection lot. An inspection lot shall correspond to each production lot.

4.5.1.3.3 Rejected lots. If an inspection lot is rejected, the supplier shall withdraw the lot and take corrective action by screening out the defective units and resubmit the lot to the test in 4.5.1.3. Resubmitted lots shall be inspected using tightened inspection in accordance with MIL-STD-105.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of those inspections in table IV, Groups A and B.

4.6.2 Inspection lot. An inspection lot shall consist of all printed wiring assemblies of the same type fabricated from the same materials and same conditions within a maximum period of 1 month, and offered for inspection at one time.

4.6.3 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table IV (Group A inspection), and shall be performed on 100 percent of all units prepared in accordance with the engineering drawings (see 3.1). Major and minor defects shall be as defined in MIL-STD-109 and 4.6.5.

4.6.3.1 Methods of examination.

4.6.3.1.1 Visual examination. The printed wiring assembly shall be examined visually for workmanship (see 3.7) and combinations of materials in accordance with the applicable requirements in table IV and the master drawings. Light intensity shall be a minimum of 100 foot-candles on the work surface. Magnification of 4X to 10X may be used to aid in inspection and questionable defects may be inspected under higher magnification.

4.6.3.1.2 Electrical parameters. The electrical parameters for the printed wiring assembly shall be checked as specified in the engineering drawings.

4.6.3.1.3 Rejection. The assembly shall be rejected if it contains any major or minor defect (see 4.6.5).

4.6.3.1.4 Rejected units. When a unit is rejected for failure to pass visual inspection, the unit may be withdrawn from the lot and corrective action taken. The unit shall be resubmitted for Group A inspection. Resubmitted lots shall be inspected using tightened inspection in accordance with MIL-STD-105.

4.6.4 Group B inspection. Group B inspection shall consist of those tests listed in table IV (Group B inspection), and shall be performed on sample units which have passed Group A inspection.

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Table IV. Quality conformance inspection.

Examination	Requirements Paragraph	Method Paragraph	Sample Plan, percent
Group A Inspection			
1. Visual Examination	4.6.3	4.6.3.1	100
a. Component Mounting	3.6.4.3 thru 3.6.4.3.11, 3.6.4	4.6.3.1.1	100
b. Printed Wiring Board	3.6.2, 3.7	4.6.3.1.1	100
c. Soldering	3.6.3.1, 3.6.3.2, 3.6.3.3, 3.6.5	4.6.3.1.1	100
d. Plating	3.6.3, 3.7	4.6.3.1.1	100
e. Cleaning	3.6.6 thru 3.6.6.4	4.6.3.1.1	100
f. Conformal Coating	3.6.8 thru 3.6.8.8	4.6.3.1.1	100
g. Sealing of Adjustable Components	3.6.9.2	4.6.3.1.1	100
h. Marking	3.6.10	4.6.3.1.1	100
2. Electrical Parameters	3.6.11	4.6.3.1.2	100
Group B Inspection	4.6.4	4.6.4.4	4.6.4.1
1. Dimensional	3.1, 3.6.1	4.6.4.4.1	4.6.4.1
2. Conformal Coating	3.6.8	4.6.4.4.2	4.6.4.1
3. Preparation for Delivery	Section 5	4.6.4.4.3	4.6.4.1

4.6.4.1 Sampling. Statistical sampling for Group B inspection shall be in accordance with MIL-STD-105 for general inspection level S-1. The AQL shall be 1 percent for major defects and 4 percent for minor defects as defined in MIL-STD-109 and 4.6.5.

4.6.4.2 Rejected lots. If an inspection lot is rejected, the supplier shall withdraw the lot and take corrective action, or screen out the defective

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units and resubmit per 4.6.4. Such lots shall be separated from new lots, and shall be clearly identified as reinspected lots. Rejected lots shall be reinspected using tightened inspection in accordance with MIL-STD-105. Each lot may be resubmitted once only.

4.6.4.3 Disposition of sample units. Samples successfully meeting Group B tests may be delivered with the accepted lot.

4.6.4.4 Methods of examination and test.

4.6.4.4.1 Dimensional examination. Printed wiring assemblies shall be examined dimensionally for compliance with the applicable drawing requirements.

4.6.4.4.2 Conformal coating examination.

a. The conformal coating thickness shall be measured on flat portions of the board, or on a test strip, in accordance with ASTM D 1005, or by any micrometers or indicators accurate to 0.0005 inches. Extremely thin paraxylene conformal coating thickness shall be measured optically.

b. The conformal coating fluorescent examination (if applicable) will be made in a dark room or hooded enclosure made for the purpose. The coating shall be viewed under ultraviolet illumination with the unaided eye. Normal or corrected 20/20 vision shall be used to examine the appearance of the conformal coating. The coating material shall be examined for evidence of bubbles, pinholes, whitish spots, blistering, wrinkling, cracking, peeling, masking or obliteration of identification markings, of discoloration of printed conductors, base materials, and evidence of foreign materials or corrosion.

4.6.4.4.3 Inspection of preparation for delivery. The sampling and inspection of the preservation packaging and interior package marking shall be in accordance with the Group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6.5 Classification of defects. Unless otherwise specified on the engineering drawing, the classification of defects for visual and dimensional examination shall be as specified herein. Some defects are listed from referenced specifications for aid only in performing inspection. The referenced specification shall be used for specified requirements.

0 - 99: Critical defect
101 - 199: Major defect
201 - 299: Minor defect

Printed Wiring Board:

101 Reduction in effective conductor pattern width by more than 35 percent, except as allowed in MIL-P-55110 determined by the

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following: A cut, crack or broken conductor extending more than 35 percent through the width of the conductor (see 3.6.2 and MIL-P-55110).

102 Lifting or wrinkling of conductor from the base laminate (see 3.6.2, 3.6.7.1, or MIL-P-55110).

103 Plating thickness is less than specified. Plating on top surface of conductors and conductor contacts does not comply with applicable engineering drawing(s) and related documents (see 3.6.3 and MIL-P-55110).

104 Unspecified removal of board material: Any visible unspecified removal of board material holes or removal of laminate about conductors (see 3.6.2 and MIL-P-55110).

105 Warp or twist of printed wiring assembly board: (see 3.6.6.3 and 4.5.1.3).

106 Any measling or crazing (prior to soldering). After soldering, measling or crazing connecting terminal areas or conductors, or covering more than three percent of the printed wiring board surface area. (Surface area is measured on one side of the printed wiring board.)

107 Delamination: Internal or external separation of layers of base material (see 3.6.3 and MIL-P-55110).

108 Peeling of plating: Plating shows evidence of peeling (see 3.6.3 and MIL-P-55110).

109 Plating on areas not specified by the engineering drawing(s) (see 3.6.3, or overlap between platings by more than 0.1 inch).

110 Excessive solder on tinned printed wiring board (see 3.6.3.3).

111 Wrong printed wiring board used (3.6.2).

112 Wrong plating used (see 3.6.3.1).

113 Interfacial connections, jumpers, etc., not in accordance with the assembly drawing (see 3.6.4.3.11, 3.6.2).

116 Inadequate cleanliness of the printed wiring assembly as evidenced by the presence of dirt, foreign matter, oil, fingerprints, corrosion, salts, flux residues and contaminants or lack of adhesion of conformal coating (see 3.6.6 and 3.7).

117 Cracks or chips in board: Cracks or chips which may tend to weaken board, other than at conductors, holes, or edges of board (see MIL-P-55110).

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201 Plating missing from one half width of conductor and the conductor is visible. Does not apply to contacts (see MIL-P-55110).

202 Pin holes, voids or small spots exposing copper conductors that exceed 20 percent but less than 35 percent of the conductor width (see 3.6.3).

203 Measling or crazing occurring on more than five percent of the quantity of the boards in a lot.

204 Reduction in effective width of conductor: Any form of scratches etc., reducing the effective width of a conductor from 20 to 35 percent (see 3.6.2).

Components and Materials:

118 Metal cased components not properly insulated from the conductor pattern or adjacent components (see 3.6.4.3.6.f).

119 Terminals: Not seated, cocked in hole. Terminal is spread outward but not curled. More than three radial cracks in rolled or swaged area. Radial cracks extending outside the rolled or swaged area, two or more radial cracks separated by less than 90° from each other, flaking or peeling of terminal plating. Any circumferential crack or split. Terminals not soldered to terminal area of printed wiring board.

121 Stress relief: Leads that have been formed too close to component body, or lead weld (see Figure 2) less than specified resulting in possible damage to component seal (see 3.6.4.3.5).

122 Component lead ends extending beyond terminal area or conductor pattern resulting in a reduction of the minimum allowable spacing between conductors (see Figure 7, 3.6.4.3.7, 3.6.4.3.8, 3.6.4.3.9).

123 Wrong parts used (see 3.6.4).

124 Components mounted in the wrong locations on the board (see 3.6.4.3).

125 Wrong orientation of polarized components (see 3.6.4.3).

126 Jumper wires not as specified on the assembly drawing (see 3.6.4.3).

127 Absence of insulation sleeving on jumper wires, when specified (see 3.6.4.3).

128 Physical damage to components (see 3.6.4.3.5, 3.6.11).

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129 Component part not properly mounted on printed circuit board (3.6.4.3).

130 Splicing of wires (see 3.6.4.3.1).

205 Component polarity markings not visible or legible (see 3.6.4.3.6.h).

207 Non-axial lead part mounting. Parts whose lead spacing is the same as that on the printed wiring board not mounted flush (in full body contact) with the board. (When spacer is provided by engineering drawing, the spacer and part combination not flush with each other and the board). Parts whose lead spacing is different from the printed wiring pattern mounted less than 0.05 or more than 0.125 inches above the board surface (see 3.6.4.3.6.c). When such parts are provided with spacers to facilitate such mounting, including spacers built onto the part, the part not flush with its spacer (if separate) and spacer not flush with board.

208 Solder extends into plated surfaces by more than 0.050 but less than 0.10 inch (see 3.6.3.1 and 3.6.3.3).

131 Soldering (see 3.6.4.3.11 and 3.6.5).

132 Reworked solder connections (see 3.6.7.2).

133 Soldering extending into plated surfaces by more than 0.10 inch (see 3.6.3.3).

134 Solder bridging between adjacent circuits, protrusions, peaks, or juts that reduce the distance between an element of one circuit and an element of an adjacent circuit or conducting material below the minimum specified herein or on the printed wiring board assembly drawing.

135 Poor wetting of solder to the basic metal, as evidenced by convex fillets, nonwetting, dewetting, cold joints, rosin joints, etc. (see 3.6.5, 3.6.7.2, 3.6.4.3.11).

136 Cracked solder joints, as evidenced by cracks, voids, gaps, blow holes, and other discontinuities in the solder joints (see 3.6.5, 3.6.7.2, 3.6.4.3.11).

Conformal Coating:

150 Absence of sealant (see 3.6.9.2).

151 Sealed adjustments free to move (3.6.9.2).

152 Blisters or bubbles in contact with, or bridging between, conductive surfaces or pin holes in the conformal coating.

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153 Cloudy areas of conformal coating covering more than 5 percent of the board surface.

154 Wrong conformal coating used (see 3.6.8).

155 Thickness and coverage of conformal coatings not in accordance with applicable drawings or specifications (see 3.6.8.5).

156 The presence of conformal coating on the adjustable portion of adjustable components (see 3.6.8.2).

157 The presence of conformal coating on electrical and mechanical mating surfaces such as connector contact points, next assembly mounting screw threads, bearing surfaces, etc. (see 3.6.8.3 and 3.6.8.4).

158 Inadequate protection of brittle components (glass, etc.), when conformal coating is used (see 3.6.4.3.6.e).

159 Leakage from oil impregnated or electrolytic components or units (see 3.6.4).

160 Scraped or otherwise abused conformal coating. Non-adherence to any surfaces to which applied. Evidence of cracks or separation from components on the printed wiring board (see 3.6.8).

169 Unauthorized repair, as evidenced by the absence of a waiver, signed by the responsible project engineer for the contracting agency, for repairs to printed wiring assemblies. This pertains to repairs but not to rework or touchup (see 3.6.7).

170 Electrical functioning of assembly not as specified in the master drawings (see 3.6.11).

171 Dimensions not as specified on the master drawings which render the printed wiring assembly unusable; will not fit in higher assembly or which affects performance (see 3.6.2).

5. PACKAGING

5.1 Level A preservation. Preserve in accordance with MIL-P-116, Method 1A-8, by individually wrapping printed wiring boards in material conforming to MIL-B-81705, Type II, or cushioned in material conforming to PPP-C-1842, Type III, Style A or B, or PPP-C-795, Class 2, or PPP-C-1752, Type VII, Class 4, or PPP-C-1797, Type II, to prevent bag puncture, and unit packed in a heat-sealed bag conforming to MIL-B-117, Type I, Class F, Style 1. Alternate cushioning materials are acceptable if certified as having physical properties equal to or better than similarly constructed material(s) covered by a government packaging specification and such materials satisfy the electrostatic decay rate requirement of MIL-B-81705. Non-corrosive non-conductive material(s) shall be applied to all exposed leads and connector pins. Lead or terminal configurations for all items shall be maintained as manufactured without

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causing loads or stresses capable of causing damage to the item. Materials used to maintain item position and lead or terminal configuration shall permit item removal without damage to the item. Bagged boards shall then be placed in a supplementary container conforming to PPP-B-1672, Type II, Style D. Container size shall be a minimum cube.

5.2 Packing.

5.2.1 Level A or B. Level A or B packing shall be in accordance with MIL-STD-2073-1, Appendix F.

5.3 Marking.

5.3.1 Level A or B. Sensitive electronic device caution labels and other markings shall be applied in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification describes effective procedures for the preparation of components for assembly to printed wiring boards, techniques for assembly and subsequent soldering, cleaning and coating processes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification
b. Title, number and revision of applicable engineering drawings
c. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
d. Whether a first article is required and how many shall be submitted (see 3.2 and 4.4.1)
e. Activity and location for first article testing (see 3.2 and 4.4)
f. Information for packaging (see Section 5).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample or a first production item, and the number of items to be tested as specified in 4.4.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Cleaning solvents. The following solvents have been found suitable for use in the cleaning process for the removal of polar and nonpolar contaminants and residues from parts, assemblies, and solder connections and will conform to the requirements of 3.4.1.1 and 3.6.6. Other products may be used with the prior approval of the design activity.

a. Chemicals:

(1) Isopropyl Alcohol (TT-I-735, Grade A)

(2) Alcohol SDA-30 (O-E-760, Type III)

(3) Trichlorotrifluoroethane (MIL-C-81302, Type 2)

(4) 1,1,1-Trichloroethane (O-T-620)

(5) Perchloroethylene (O-T-236)

(6) Water, Deionized (ASTM D 1193, Grade IV)

b. Solutions:

(1) 35 percent weight, Isopropyl Alcohol (TT-I-735, Grade A)

65 percent weight, Trichlorotrifluoroethane (MIL-C-81302, Type 2)

(2) 15 percent weight, Isopropyl Alcohol (TT-I-735, Grade A)

85 percent weight, Trichlorotrifluoroethane (MIL-C-81302, Type 2)

(3) 35 percent weight, Alcohol SDA-30 (O-E-760, Class A, Grade 3)

65 percent weight, Trichlorotrifluoroethane (MIL-C-81302, Type 2)

(4) 14.6 percent weight, Isopropanol (TT-I-735, 2 Propanol, Grade A)

85.4 percent weight, Tetrachlorodifluoroethane (CCl_2 F-CCl_2 F)

(5) 28.4 percent weight, Isopropanol (TT-I-735, 2 Propanol, Grade A)

71.6 percent weight, Tetrachlorodifluoroethane (CCl_2 F-CCl_2 F)

(6) 4 percent weight, Ethyl Alcohol SDA-30 (O-E-760, Class A, Grade 3)

96 percent weight, Trichlorotrifluoroethane (MIL-C-81302, Type 2)

NOTE: Due to the many manufacturing processes and materials used to clean printed wiring board assemblies after soldering, it is the responsibility of the producing facility to determine the suitability of the solvents selected for the cleaning processes.

CAUTION: Trichloroethane and other chlorinated hydrocarbon fluids should be used with caution and in well-ventilated rooms.

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6.5 Subject term (keyword) listing.

Component leads
Component mounting
Lead termination
Printed circuit boards
Soldering electronic components

6.6 Metrication. Whenever inch/pound dimensions are used in this document, metric equivalents in accordance with FED-STD-376 are acceptable.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Project No. 5999-A096

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-46843C(MI)	Printed Wiring Assemblies, Production of

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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